Retail Sales Performance Project Report

Data Cleaning, Exploratory Analysis, Statistical Testing & Power BI Dashboard

Dataset: 100,000 retail orders | India | Jan 2019 – Dec 2023
Tools: Python (pandas, SciPy) for analysis | Power BI for dashboarding

Executive Summary

This project analyzes five years of retail transaction data (2019–2023) covering 100,000 orders across 10 Indian states, 6 product categories, and 100,000 unique customers. The workflow spans raw data cleaning and feature engineering in Python, exploratory and statistical analysis, and an interactive two-page Power BI dashboard for ongoing business monitoring.

The headline finding is stability rather than growth: revenue held almost perfectly flat across all five years (year-over-year change never exceeded 0.3% in either direction), and a formal statistical test found no significant difference in sales performance across states. The more actionable finding is around discounting: profit margin declines in an almost perfectly linear, consistent pattern as discount level increases — and this pattern holds uniformly across every product category, suggesting a single company-wide discounting policy is likely driving the effect rather than category-specific dynamics.

Metric	Value
Total Revenue	$2.51bn
Total Profit	$375.53M
Overall Profit Margin	14.97%
Total Orders	100,000
Unique Customers	100,000
Average Order Value	$25,084

1. Methodology

1.1 Data Cleaning & Feature Engineering (Python / pandas)

The raw dataset was cleaned and enriched in a Jupyter notebook before being loaded into Power BI. Key steps:

- Standardized column names (lowercase, underscores, stripped whitespace/symbols)
- Converted order_date, ship_date, sales_date, and date_of_birth to proper datetime types
- Derived customer_age from date_of_birth, correctly accounting for whether the birthday has occurred yet this year
- Engineered time features: month, month_name, order_year, and order_month_year for trend analysis
- Calculated shipping_days (ship_date – order_date) as an operational metric
- Calculated profit_margin_percent per order
- Binned discount into a discount_band category (0–10%, 10–20%, ... 40–50%)
- Binned customer_age into age_group segments (<20, 20-30, ... 60+)
- Built an RFM-style customer_segment (Best Customers / Loyal-High Value / At Risk / Lost-Low Value) using recency and monetary value quartiles
- Validated data quality: checked for nulls, duplicates, and invalid ages (under 18)

1.2 Statistical Testing

A one-way ANOVA test was run to check whether average sales differ significantly across the 10 states, testing the null hypothesis that all states have equal mean sales.

1.3 Dashboard

A two-page Power BI dashboard was built on top of the cleaned dataset: an Executive Overview page (KPIs, monthly trend, regional and category breakdowns, customer segmentation) and a Probability Analysis page (discount impact on margin, profit by region, age group performance). Custom DAX measures were built for time-intelligence (YoY growth), discount analysis, and customer segmentation, styled with a custom Power BI theme for a consistent visual identity across both pages.

2. Key Findings

2.1 Revenue is flat, not growing

Year-over-year revenue barely moved across the entire five-year window: 2019 ($502.2M) → 2020 (–0.28%) → 2021 (+0.10%) → 2022 (+0.13%) → 2023 (+0.06%). This is a business running at a steady state rather than growing or declining — worth flagging to stakeholders, since flat revenue over 5 years usually warrants investigating whether it reflects market saturation, a pricing ceiling, or a lack of growth initiatives during the period.

2.2 State-level differences are not statistically significant

An ANOVA test across all 10 states returned F = 1.60, p = 0.108. Since p > 0.05, we cannot reject the null hypothesis that average sales are equal across states. In practical terms: although Rajasthan, Maharashtra, and Punjab nominally rank highest by profit, the gap between the best- and worst-performing states is small enough to plausibly be random variation rather than a real regional effect. This is a useful check against over-interpreting small differences in a regional performance ranking.

2.3 Discounting has a strong, remarkably consistent effect on margin

Profit margin declines steadily as discount level increases, dropping from roughly 19% average margin at the 0–10% discount band to roughly 11% at the 40–50% band — an ~8 percentage-point decline. What stands out is how uniform this pattern is: it holds almost identically across all six product categories, with each category showing nearly the same margin at each discount band. This strongly suggests a single, company-wide discounting mechanism is driving margin erosion, rather than the effect varying meaningfully by product type.

2.4 Revenue is not concentrated in a small customer group

The top 20% of customers by spend account for only about 36% of total revenue — well short of the classic 80/20 Pareto pattern often assumed in retail. Revenue here is fairly evenly distributed across the customer base rather than concentrated in a small group of power buyers, which has implications for retention strategy: losing any single high-value customer is less risky here than in a business with a true Pareto concentration, but it also means broad-based retention efforts likely matter more than VIP-focused ones.

2.5 Customer segment concentration (RFM)

By the RFM-based segmentation, Loyal/High-Value customers drive 48.6% of revenue and Best Customers drive another 29.6% — together, 78% of revenue comes from these two healthy segments. At-Risk customers still contribute 20.3% of revenue, representing a meaningful pool worth targeted retention effort before they churn into the Lost/Low-Value segment (currently just 1.6% of revenue).

2.6 Age group spending shows a U-shaped pattern

The under-20 and 60+ age groups are the two highest-spending segments ($478M and $573M respectively), notably higher than the middle age brackets (20-30 through 50-60), which each sit close to $362-367M. This U-shape is worth further investigation — it may reflect different purchasing occasions (e.g., family/gift purchases at the youngest bracket, or larger basket sizes among older, higher-income customers).

3. Recommendations

- **Review discounting policy centrally** rather than per-category, since the margin impact is uniform across categories — a single company-wide discount cap or tiered approval process would likely be more effective than category-specific rules.
- **Investigate the flat revenue trend** with the broader business team: is this a deliberate steady-state period, or a signal that new growth levers (channels, categories, geographies) are needed?
- **Prioritize retention campaigns for the At-Risk segment** (20.3% of revenue) before those customers churn into Lost/Low-Value — this is a larger and more recoverable opportunity than acquiring new customers.
- **Do not over-invest in state-specific strategy** based on the current ranking alone, given the ANOVA result — consider re-testing with a longer time window or investigating specific high/low performing cities within states instead.
- **Explore the U-shaped age group pattern** with qualitative research (e.g., customer surveys) to understand whether under-20 and 60+ spending is driven by different purchase motivations, which could inform targeted marketing.

4. Project Structure

This repository contains the full workflow from raw data to dashboard: the Jupyter notebook (`Retail_Sales_Performance.ipynb`) documents every cleaning and analysis step described in this report; the Power BI file (`Retail_Sales_Dashboard.pbix`) contains the interactive two-page dashboard; and the cleaned dataset (`cleaned_retail_sales.csv`) is the direct output of the notebook and the data source for the dashboard. See the README for the full folder structure and setup instructions.

Prepared as a portfolio analytics project. Dataset is synthetic/cleaned retail data used for demonstration purposes.